

Multi-Criteria Assessment of Antimony Exploration Potential in Alaska

Cross-referencing mining occurrences, historical grades, geochemical anomalies, and geological context

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Summary

Antimony is a metalloid used notably in alloys, flame retardants, and certain electronic and military applications, which makes it a substance closely tracked in strategic mineral resource inventories. This report proposes a multi-criteria screening analysis aimed at identifying and prioritizing sites of potential interest for antimony exploration across the territory of Alaska. The question examined is as follows: how can heterogeneous historical information, drawn from old mining archives, be combined with more recent regional geochemical data to establish a comparative ranking of candidate sites for further exploration?

The study draws on three main sources. The Alaska Resource Data File (ARDF), maintained by the United States Geological Survey (USGS), records mineral occurrences and former sites associated with antimony. The Alaska Geochemical Database, version 4 (AGDB4), provides antimony concentrations measured in stream-sediment samples at the state scale. The generalized geologic map of Alaska (USGS Scientific Investigations Map 3340) provides descriptive geological context. Data processing was carried out using Python and the pandas library for tabular manipulation, and QGIS for spatial operations and map production.

The approach consisted of filtering the ARDF database to isolate records mentioning antimony, consolidating the available grade and tonnage information for a subset of thirteen documented sites, extracting and classifying antimony geochemical anomalies in stream sediments, and then spatially comparing these anomalies with the location of the thirteen sites. A final score, combining a site-specific screening score and a score reflecting regional geochemical support, made it possible to sort the sites into three final priority classes.

Of the 7,720 records in the ARDF database, 887 occurrences were retained after filtering and harmonization. Among the thirteen sites for which grade data could be consolidated, seven achieve a high final priority, five a medium final priority, and one a low final priority. The Scrafford (Treasure Creek) and Pringle Bench (Jones & Boyle) sites obtain the highest score, tied. Analysis of the 129,074 stream-sediment samples made it possible to identify 3,110 antimony geochemical anomalies. Spatial analysis shows that several of the documented sites are associated with nearby geochemical anomalies, which provides an element of consistency between the mining archives and the regional geochemical data.

This study nonetheless presents important limitations that must be kept in mind. It is an exploratory screening analysis based on heterogeneous historical data and regional geochemical coverage; it does not constitute in any way a resource or reserve estimate, a feasibility study, an economic assessment, or a field geological validation. The proposed ranking only serves to guide subsequent verification work and should not be interpreted as a definitive hierarchy of the mining potential of the sites considered.

Keywords: antimony; Alaska; mineral exploration; multi-criteria analysis; geochemical anomalies; geographic information systems.

Introduction

Antimony, chemical symbol Sb, is a metalloid used mainly as a hardening agent in certain alloys, as a component of flame retardants, and in several electronic and defense applications. Its global production remains concentrated in a small number of countries, which leads several administrations to classify it among mineral substances considered critical or strategic. In this context, identifying areas or sites with exploration potential outside the main current production zones is of interest that extends beyond the purely geological framework.

Alaska is a territory in which numerous antimony occurrences have been recorded for more than a century, without however having been the subject of a systematic comparative synthesis. The available information comes from heterogeneous sources: historical descriptions of mines and prospects, regional geochemical measurements collected at different periods, and generalized geological mapping. This heterogeneity makes direct comparison between sites difficult and justifies the use of a multi-criteria approach, which makes it possible to combine several types of information without artificially merging them into a single indicator lacking methodological justification.

The research question structuring this report can be formulated as follows: based on the historical mining occurrences and geochemical data available for Alaska, which sites present the strongest relative interest for further antimony exploration, and on what basis can this ranking be established transparently?

The general objective of the study is to propose a comparative ranking of sites, based on explicit and verifiable criteria. The specific objectives consist of identifying antimony occurrences in the ARDF database, consolidating the available grade and tonnage information for the best-documented sites, characterizing regional antimony geochemical anomalies, spatially comparing these anomalies with the recorded sites, and then combining these elements into a final screening score.

The approach followed relies on programmatic data processing using Python, complemented by spatial operations carried out in QGIS. The report is organized into eight main parts. The study context presents the basic concepts necessary to understand the subject. The data and sources used are then detailed, followed by a description of the processing applied. The multi-criteria assessment methodology is then presented, before the presentation of results, organized into five sub-sections corresponding to occurrences, grades, geochemical anomalies, geochemical support of sites, and the final ranking. A discussion, recommendations, a section devoted to the study's limitations, and a conclusion complete the report, followed by the bibliography and appendices.

1. Study Context

This first part presents the concepts necessary to understand the rest of the report, without claiming to constitute a general introduction to geology or the mining industry. Each concept is presented in relation to its use in the analysis that follows.

1.1. Antimony and Stibnite

Antimony is a metalloid chemical element, meaning it exhibits properties of both metals and non-metals. In nature, it rarely occurs in pure form and is most often found combined with other elements as minerals. The main sulfide mineral of antimony is stibnite, a compound with the formula Sb_2S_3 , which has historically been the main source of extraction of this metal. The presence of stibnite in a site description can thus signal antimony mineralization, including in cases where antimony is not properly recorded in a database's main fields. For this reason, mentions of stibnite were retained as a supplementary criterion for identifying sites in this study.

1.2. Occurrence, Prospect, and Mine

Geoscience databases generally use three categories to describe a site where mineralization has been observed. An occurrence corresponds to a site where a mineral presence has been identified, without extensive exploration work necessarily having been carried out there. A prospect designates a site that has been the subject of more advanced exploration, for example drilling or reconnaissance work, without exploitation having been undertaken. A mine corresponds to a site that has experienced mining work or exploitation at some point in its history, without this meaning that the site is still active today. These three categories therefore do not prejudge a site's current potential, but provide information on the intensity of work previously conducted there.

1.3. Grade, Tonnage, and Stream-Sediment Sample

The grade of a sample designates the proportion of the element considered, here antimony, in that material, most often expressed as a percentage when dealing with relatively high grades. Tonnage corresponds to the quantity of material documented for a site, expressed in units that may vary from one source to another. A stream-sediment sample is a sample of fine materials accumulated at the bottom of a watercourse; these sediments originate from the erosion of rocks and soils located upstream of the sampling point, across the entire watershed concerned. Chemical analysis of these sediments makes it possible to detect abnormally high concentrations of certain elements, which may reflect the presence of a mineralized source within the watershed, without however making it possible to precisely locate that source.

1.4. Geochemical Anomaly and the ppm Unit of Measurement

A geochemical anomaly corresponds to a measured concentration of an element that lies significantly above the values usually observed in a given region. Antimony concentrations in stream sediments are expressed in parts per million, or ppm, a unit that corresponds approximately to one unit of the element

considered per million units of material analyzed. A geochemical anomaly does not, however, constitute direct proof of a deposit: it may result from the natural geology of the watershed, erosion and sediment transport phenomena, the size of the sampled watershed, the analytical limits of measurement methods, or the presence of former mining works that locally enriched the sediments. Interpretation of an anomaly must therefore remain cautious.

1.5. Multi-Criteria Analysis and Levels of Subsurface Knowledge

Multi-criteria analysis is a method that consists of combining several indicators of different natures in order to establish a ranking or comparison among several elements, here mining sites. This approach makes it possible to take several dimensions of information into account simultaneously without having to choose a single criterion, at the cost of an explicit choice of combination rules, which must be presented transparently.

Finally, several notions used throughout this report must be distinguished and should not be confused. Exploration priority, as established in this study, indicates a relative interest in further verification work. Geological potential refers to a qualitative appraisal of a context favorable to mineralization. A mineral resource designates a concentration of material whose existence, quantity, and grade have been estimated with a certain level of confidence according to recognized methods, whereas a mineral reserve corresponds to the portion of that resource deemed extractable under given technical and economic conditions. Economic viability, finally, depends on numerous factors that fall well beyond the scope of this analysis, such as operating costs, site accessibility, or market prices. None of the data used in this report makes it possible to establish a resource, a reserve, or economic viability within the meaning of these definitions.

2. Data and Sources

This part presents the data sources used in the study, their content, and their respective role in the analysis. Table 1 provides a summary before each source is detailed.

Table 1. Sources of Data Used in the Study

Source	Data Type	Count	Main Variables	Role in the Analysis	Main Limitations
ARDF (USGS)	Mineral occurrences	7,720 initial records; 887 retained	Site name, substances, site type, production status	Identification and characterization of sites	Sometimes old or incomplete descriptions
AGDB4 (USGS)	Stream-sediment geochemistry	129,074 samples	Sb concentration (ppm)	Detection of regional geochemical anomalies	Uneven spatial coverage, values below detection limit
Consolidated grades	Historical grade and tonnage data	18 observations, 13 sites	Mean Sb grade, documented tonnage	Comparison of documented sites	Heterogeneous measurement contexts, variable units
Generalized geologic map (USGS SIM 3340)	Regional geology	Polygon map layer	Unit code, age, geological context	Descriptive and interpretive variable	Regional scale, not integrated into the numerical score
Administrative boundaries and cities	Reference base layer	Vector layers	Borders, city locations	Geographic reference on maps	No geological information
OpenStreetMap	Map background	Map tiles	Road network, hydrography, relief	Facilitates location on maps	Provides no geological or mining information

Source: prepared by the author from the methodological information provided for this study.

2.1. The Alaska Resource Data File (ARDF)

The Alaska Resource Data File, abbreviated ARDF, is a database maintained by the United States Geological Survey that records mineral occurrences and former mining sites in the State of Alaska. For each site, it records in particular the site name, the main and secondary associated substances, the minerals mentioned in the descriptions, the site type (occurrence, prospect, or mine), and the production status. The initial ARDF database includes 7,720 records, across all substance types. The filtering carried out for this study made it possible to isolate records related to antimony, according to a procedure detailed in Part 3.

2.2. The Alaska Geochemical Database, Version 4 (AGDB4)

The Alaska Geochemical Database, version 4, abbreviated AGDB4, compiles the results of geochemical analyses carried out on samples collected in Alaska, including stream-sediment samples. For antimony, this database provides concentrations expressed in ppm for each sample analyzed. The extraction carried out for this study covers 129,074 stream-sediment samples, the processing of which is detailed in Part 3.

2.3. Historical Grade and Tonnage Data

A consolidated database was built from the historical descriptions available in the ARDF for sites where grade or tonnage information was mentioned. This database comprises 18 observations corresponding to

13 distinct sites. This information comes from documents and descriptions of varying nature and period, which implies significant heterogeneity, detailed in Part 3.

2.4. Generalized Geologic Map of Alaska

The regional geological context used in this report comes from the generalized geologic map of Alaska, published by the United States Geological Survey under reference Scientific Investigations Map 3340. The layer used in QGIS, named AKStategeolpoly_generalized, provides for each geological polygon a unit code, a symbol, a unit name, a geological age, and a descriptive geological context. This information is used for descriptive and interpretive purposes; it does not enter as a numerical component of the final score used to rank the sites.

2.5. Administrative Boundaries and Map Background

The maps presented in this report also rely on a layer representing Alaska's boundaries, a layer representing certain reference cities, as well as an OpenStreetMap background. This background does not constitute a source of geological or mining information; it serves only to facilitate the location of sites and anomalies across the territory.

3. Data Preparation and Processing

This part describes the processing applied to the raw data to produce the tables and maps presented in this report. Emphasis is placed on justifying each step rather than on the technical detail of its execution.

3.1. Processing of the ARDF Database

The initial ARDF database, comprising 7,720 records, was filtered in order to isolate records related to antimony. This filtering made it possible to identify 289 records where antimony is indicated as the main substance, 573 records where antimony is indicated as a secondary substance, and 512 records containing a mention of stibnite in their description. These three sets partially overlap, as the same site may appear under several filtering criteria.

After harmonizing site names and removing duplicates among these three sets, 887 records were retained in total. The final breakdown distinguishes 289 occurrences where antimony appears as the main substance, 569 occurrences where antimony appears as a secondary substance, and 29 occurrences retained solely due to a mention of stibnite in their description, without antimony being recorded in the corresponding main fields.

3.2. Consolidation of Grades and Tonnages

The historical descriptions associated with certain ARDF sites mention grade or tonnage information. These mentions were extracted and compiled into a consolidated database of 18 observations covering 13 sites. The quality of this information is variable: 14 observations include both a grade and a tonnage, 2 observations include only a grade, 1 observation includes only a tonnage, and 1 observation was judged insufficiently documented to be reliably used.

Particular attention was paid to the heterogeneous nature of this information. A grade mentioned in a historical description may correspond to a spot sample, an average production grade, an old estimate, a mineralized interval, a hand-sorted ore grade, or a partial grade mentioned in an old report. These grades were therefore not treated as comparable measurements obtained under a single protocol. Likewise, the documented tonnages are expressed in different units, notably tons, short tons, or kilograms depending on the site. Values expressed in kilograms or short tons were converted into metric tons before assigning the tonnage score, using a precise conversion factor; values expressed under the sole generic mention of “tons,” however, were not converted, since the exact nature of this unit is not confirmed in the files provided, which limits the full comparability of tonnages between sites.

3.3. Extraction and Classification of Stream Sediments

The extraction carried out in the AGDB4 database covered 129,074 stream-sediment samples. Of these, 28,968 include a measured antimony value, 91,212 correspond to results below the detection limit of the analytical method used, and 8,894 records contain no usable value. For measured values, the median is 1.4 ppm, the 95th percentile is 14.8 ppm, the 98th percentile is 50 ppm, and the maximum observed value is approximately 100,000.11 ppm.

A median corresponds to the value that divides an ordered set of observations into two equal halves. A percentile corresponds to the value below which a given percentage of observations falls: the 95th percentile thus corresponds to a value exceeded by about 5% of measured observations, and the 98th percentile to a value exceeded by about 2% of observations. The detection limit refers to the threshold below which an analytical method can no longer reliably distinguish a measured concentration from the absence of a signal; results below this threshold are reported as such rather than as a precise numerical value.

From these measured values, 3,110 geochemical anomalies were extracted and classified into three levels: 1,660 moderate anomalies, 741 high anomalies, and 709 very high anomalies. For mapping purposes only, 23 values above 5,000 ppm were visually capped at this threshold, to prevent a small number of extreme values from making the map difficult to read. This capping is strictly graphical and does not modify the original values retained in the data.

3.4. Spatial Comparison Between Sites and Anomalies

A spatial operation made it possible to compare the location of the thirteen documented sites with the presence and intensity of nearby geochemical anomalies. This comparison led to classifying each site according to a level of geochemical support, presented in Section 4.

3.5. Joining with Geology and Construction of the Final Ranking

The sites were matched with the generalized geological layer, which made it possible to associate each with a unit code, a unit name, a geological age, and a descriptive geological context. This information was retained for descriptive purposes. The final ranking of sites was then built by combining the site-specific screening score and the geochemical support score, according to the methodology presented in Part 4.

4. Multi-Criteria Assessment Methodology

This part presents the construction of the final ranking of the thirteen documented sites. The calculation rules presented below were recovered from the Python scripts provided by the author (05_build_site_screening.py, the geochemical proximity script, and the final ranking script) and are reproduced here in English, without reproducing the code. The scoring system is thus explicit and reproducible; it nonetheless remains based on threshold and weighting choices specific to this study, discussed in Section 4.6.

4.1. Steps of the Approach

The construction of the ranking relies on a succession of steps applied to the full set of sites considered. The first step consists of selecting antimony occurrences in the ARDF database. The second step characterizes the status of each site, notably its type and production status. The third step consolidates the available grades and tonnages for the documented sites and assigns them a screening score. The fourth step extracts regional antimony geochemical anomalies. The fifth step spatially compares each site with nearby anomalies and derives a geochemical score from this. The sixth step calculates the final score by adding the screening score and the geochemical score. The seventh step sorts the sites into three final priority levels. The eighth step, finally, associates each site with a descriptive geological context, which does not enter into the numerical calculation of the score.

4.2. Screening Score

The screening score of a site results from the addition of four components: a grade score, a tonnage score, a documentary quality score, and a quantitative completeness score. Table 2 presents the calculation rule for each of these components.

Table 2. Screening Score Calculation Rules

Component	Variable Used	Calculation Rule	Points
Grade score	Mean Sb grade (%)	Grade $\geq$ 50%	5
		30% $\leq$ grade < 50%	4
		15% $\leq$ grade < 30%	3
		10% $\leq$ grade < 15%	2
		Grade < 10% (but recorded)	1
		Grade not recorded	0
Tonnage score	Documented tonnage, harmonized to metric tons where the unit allows	Tonnage $\geq$ 5,000	5
		1,000 $\leq$ tonnage < 5,000	4
		100 $\leq$ tonnage < 1,000	3
		10 $\leq$ tonnage < 100	2

Component	Variable Used	Calculation Rule	Points
		Tonnage < 10 (but recorded)	1
		Tonnage not recorded	0
Documentary quality score	Nature of the grade source (maximum value retained per site)	Mean resource grade	5
		Average or production grade	4
		Hand-sorted ore grade or historical estimate	3
		Spot-sample grade or contained metal	2
		Context not recorded	1
Quantitative completeness score	Presence of a grade and a tonnage	1 point if grade recorded, 1 point if tonnage recorded	0 to 2

Source: prepared by the author from script 05_build_site_screening.py provided for this study.

The documentary quality score retains, for each site, the highest value among the set of historical observations attached to that site, rather than an average or the most recent value. This choice favors the best information available for each site; it is a methodological choice specific to this study, which does not necessarily constitute the only possible method for aggregating multiple observations. A site's screening score corresponds to the sum of its four components, which explains why the observed values in the data range from 1, for the least documented site, to 14, for the best-documented sites.

A screening score of 14 or higher corresponds to a high screening priority; a score between 10 and 13 corresponds to a medium priority; a score below 10 corresponds to a low priority.

The script initially assigns tonnage thresholds to the raw numerical value, without distinguishing the unit of measurement. For this version of the report, this value was harmonized to metric tons where the declared unit reliably allowed it: tonnages expressed in kilograms were converted using the factor 0.001, and those expressed in short tons using the factor 0.90718474. However, the majority of sites mention only a generic “tons” unit, whose exact nature (US short ton, metric ton, or long ton) is not confirmed in the files provided; these values were therefore not converted, in order to avoid an arbitrary equivalence, and their tonnage score remains calculated on the raw value. The comparability of tonnages between sites therefore remains incomplete, and this limitation is revisited in Section 4.6.

4.3. Geochemical Score

A site's geochemical score combines three components based on antimony anomalies detected in stream sediments. The distance to the nearest anomaly is calculated from the full set of geochemical anomalies, across all classes, using the Haversine formula, which gives the great-circle distance between two points identified by their geographic coordinates. The number of anomalies considered within a 25-kilometer radius, however, covers only very high-class anomalies. The maximum concentration measured within this same 25-kilometer radius is calculated across all anomalies present within that radius, without class restriction.

Table 3. Geochemical Score Calculation Rules

Component	Calculation Rule	Points
Distance to the nearest anomaly (all classes)	Distance $\leq$ 0.5 km	4
	0.5 km < distance $\leq$ 2 km	3
	2 km < distance $\leq$ 5 km	2
	5 km < distance $\leq$ 10 km	1
	Distance > 10 km	0
Number of very high anomalies (25 km radius)	30 anomalies or more	4
	15 to 29 anomalies	3
	5 to 14 anomalies	2
	1 to 4 anomalies	1
	No very high anomaly	0
Maximum Sb concentration, all classes (25 km radius)	Concentration $\geq$ 5,000 ppm	3
	1,000 ppm $\leq$ concentration < 5,000 ppm	2
	200 ppm $\leq$ concentration < 1,000 ppm	1
	Concentration < 200 ppm	0

Source: prepared by the author from the geochemical proximity script provided for this study.

A site's geochemical score corresponds to the sum of these three components, which explains why the observed values in the data range from 4 to 11 across the thirteen sites studied. A geochemical score of 9 or higher corresponds to strong support; a score between 6 and 8 corresponds to moderate support; a score below 6 corresponds to limited support.

The final ranking file provided for this study retains only the aggregated geochemical score for each site; it does not retain the exact distance to the nearest anomaly, the number of very high anomalies, or the maximum concentration used in this calculation. It is therefore not possible to present, in this report, the breakdown of the geochemical score specific to each site, whereas this breakdown is possible for the screening score and appears in Appendix 2.

4.4. Final Score and Priority Classes

The final score assigned to each site results from the following combination:

$$\text{Final score} = \text{screening score} + \text{geochemical score}$$

The screening score, the geochemical score, and the final score each rely on their own thresholds, which must not be confused. Table 3 bis presents them side by side.

Table 3 bis. Thresholds of the Three Priority Systems

Indicator	High Level	Medium Level	Low Level
Screening score	≥ 14	10 to 13	< 10
Geochemical score (support)	≥ 9	6 to 8	< 6
Final score (priority)	≥ 20	14 to 19	< 14

Source: prepared by the author from the functions `priority_class` (05_build_site_screening.py), `geochemical_priority` (geochemical_proximity script), and `classify_final_priority` (final ranking script).

4.5. Geological Context

The geological context associated with each site is determined from the name and age of the corresponding geological unit, through a keyword reading of these two fields. The presence of terms such as schist, phyllite, metasedimentary, or metamorphic classifies the site as a metamorphic or metasedimentary context. The presence of terms such as volcanic, andesite, or basalt classifies it as a volcanic context. The presence of terms such as granite or plutonic classifies it as an intrusive context. The presence of terms such as sedimentary, limestone, or sandstone classifies it as a sedimentary context. In the absence of a match with any of these keyword groups, the site is classified as an other or mixed context. This classification is purely descriptive and adds no points to the final score, as no scientific weighting has been defined by the author of the study for the geological units concerned. It explains why the thirteen sites studied fall into only two of these categories, none of the geological units associated with these sites containing keywords specific to volcanic or sedimentary contexts.

4.6. Methodological Limitations of the Scoring System

The fact that the scoring system is now explicit and reproducible from the scripts does not mean it is free of limitations, which should be gathered here.

- sensitivity of the ranking to the thresholds retained: small changes to these thresholds could change the rank or class of certain sites, particularly those whose score lies close to a class boundary;
- weightings chosen by the author of the study, without external scientific justification demonstrating that this is the only possible or most relevant combination;
- absence of a systematic sensitivity analysis to measure the exact extent of the effect of these choices on the final ranking;
- use of the maximum documentary quality value per site, a methodological choice that favors the best available source rather than an average or the most recent source;
- persistent ambiguity of the “tons” unit for the majority of sites, which could not be reliably converted and which limits the comparability of tonnages despite the conversion of the two sites expressed in kilograms and short tons;
- uneven geochemical coverage across sectors of the territory, likely to influence the geochemical score independently of a site's actual potential;
- regional character of the geological and geochemical data used, which does not account for local variations specific to each site;
- absence of external validation of the scoring system as a whole, which has been neither compared to an independent dataset nor submitted to the appraisal of a specialized third party.

5. Results

This part presents the results of the analysis, organized into five sub-sections corresponding respectively to the recorded antimony occurrences, the documented grades and tonnages, the geochemical anomalies, the geochemical support of sites, and the final ranking.

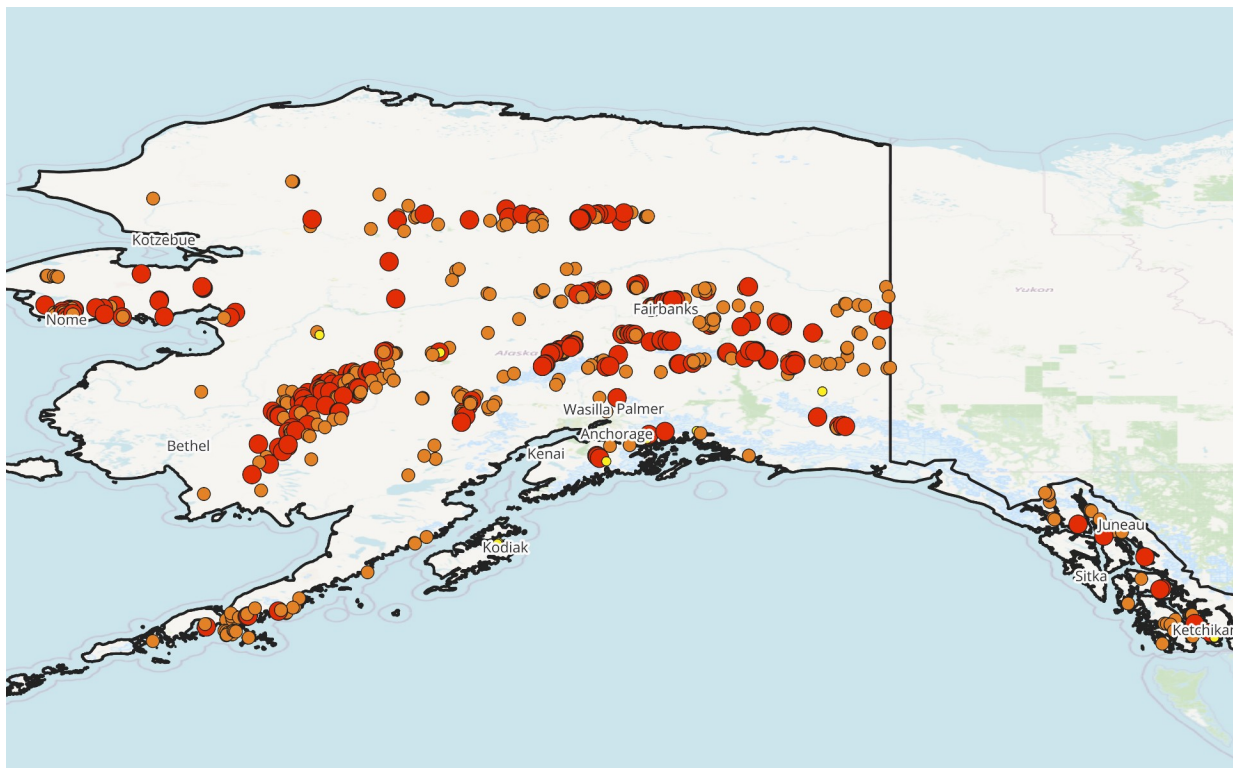
5.1. Recorded Antimony Occurrences

The filtering of the ARDF database, described in Part 3, made it possible to go from 7,720 initial records to 887 occurrences retained in relation to antimony. This reduction corresponds to the application of three successive filtering criteria, whose results partially overlap: the mention of antimony as the main substance, its mention as a secondary substance, and the presence of the term stibnite in the site description.

Among the 887 occurrences retained, 289 include antimony as the main substance, or about 32.6% of the total; 569 include antimony as a secondary substance, or about 64.2%; and 29 were retained solely because of a mention of stibnite, or about 3.3%. Regarding site type, 422 records are classified as prospects, or about 47.6% of the total; 234 are classified as occurrences, or about 26.4%; and 231 are classified as mines, or about 26.0%. This distribution indicates that a significant share of the recorded sites has only been subject to limited exploration, while about a quarter of the sites have experienced actual mining work at some point in their history, without this presuming their current status.

Regarding production status, 584 sites have an unknown status, 229 sites have documented production mentioned, and 74 sites have an indeterminate status. The mention of historical production does not necessarily mean that significant production took place, that the site remains workable, that the historical information would meet current resource-estimation standards, or that the site would show economic viability in the current context.

Map 1 presents the spatial distribution of these 887 occurrences across the territory of Alaska.



Map 1. Distribution of Recorded Antimony Occurrences in Alaska

Reading note: the map distinguishes three categories of occurrences according to how antimony is mentioned in the ARDF database: sites where it constitutes the main substance, those where it constitutes a secondary substance, and those retained solely because of a mention of stibnite in their description. These three categories are represented by symbols of different shades, whose exact correspondence with each category is indicated in the legend of the original map. Visual inspection of the map suggests a greater concentration of occurrences in several sectors of central Alaska and around Nome. This concentration may reflect local geology, the historical intensity of exploration, sector accessibility, or data availability; these factors cannot be distinguished from one another based on this map alone. An area with few points should not be interpreted as lacking antimony; it may simply correspond to a sector less explored historically.

Source: prepared by the author from the USGS Alaska Resource Data File. Map background: OpenStreetMap.

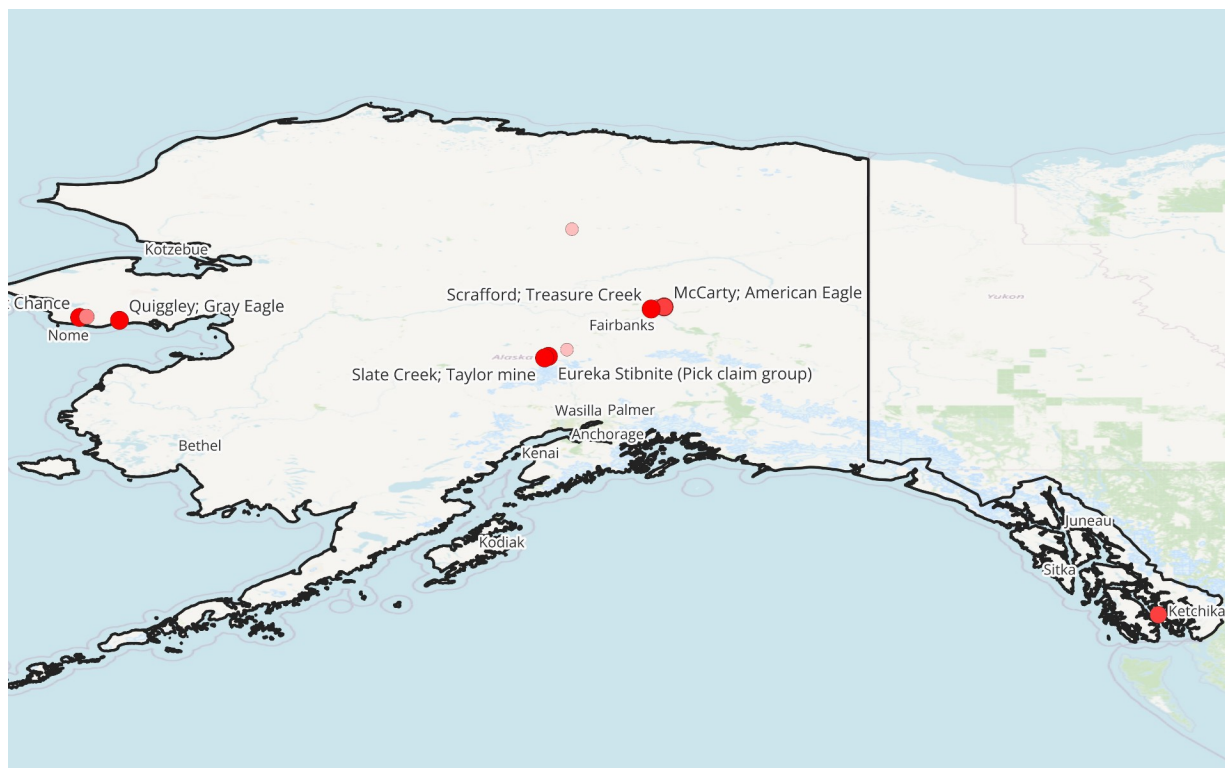
5.2. Documented Grades and Tonnages

For the thirteen sites with consolidated grade or tonnage information, documentation quality remains variable. Of the 18 observations compiled, 14 include both a grade and a tonnage, 2 include only a grade, 1 includes only a tonnage, and 1 was judged insufficiently documented. The grades available in the ranking file range from about 10.0% Sb for the Breen West (Bison) site to about 63.7% Sb for the Quiggley (Gray Eagle) site, with the Upper Ridge site having neither grade nor tonnage documented in the data provided.

These grade values cannot be compared as though they all resulted from an identical measurement protocol. Depending on the historical descriptions from which they originate, they may correspond to a spot sample, an average production grade, an old estimate, a mineralized interval, or a hand-sorted ore grade. Likewise, the documented tonnages are expressed in different units depending on the site, notably

tons, short tons, or kilograms. Values expressed in kilograms or short tons could be converted into metric tons using a precise conversion factor, but most sites mention only a generic “tons” unit, whose exact nature is not confirmed in the files provided; these latter values were therefore not converted, and the tonnages in the report should not be considered fully harmonized with one another.

Map 2 presents the spatial distribution of these thirteen sites, classified according to their documented mean antimony grade.



Map 2. Documented Antimony Grades for the Retained Sites

Reading note: the symbols classify the thirteen documented sites into four classes of mean antimony grade: from 0 to 13% Sb, from 13 to 39% Sb, from 39 to 49% Sb, and from 49 to 64% Sb. Symbol color and size represent these classes, with the largest and darkest symbols corresponding to the highest grades. These grades come from historical documents and different sampling contexts; they therefore do not constitute directly comparable resource estimates. A high grade demonstrates neither a large tonnage, nor the existence of a resource in the mining sense of the term, nor economic profitability.

*Source: prepared by the author from grade data consolidated from USGS Alaska Resource Data File records and descriptions.
Map background: OpenStreetMap.*

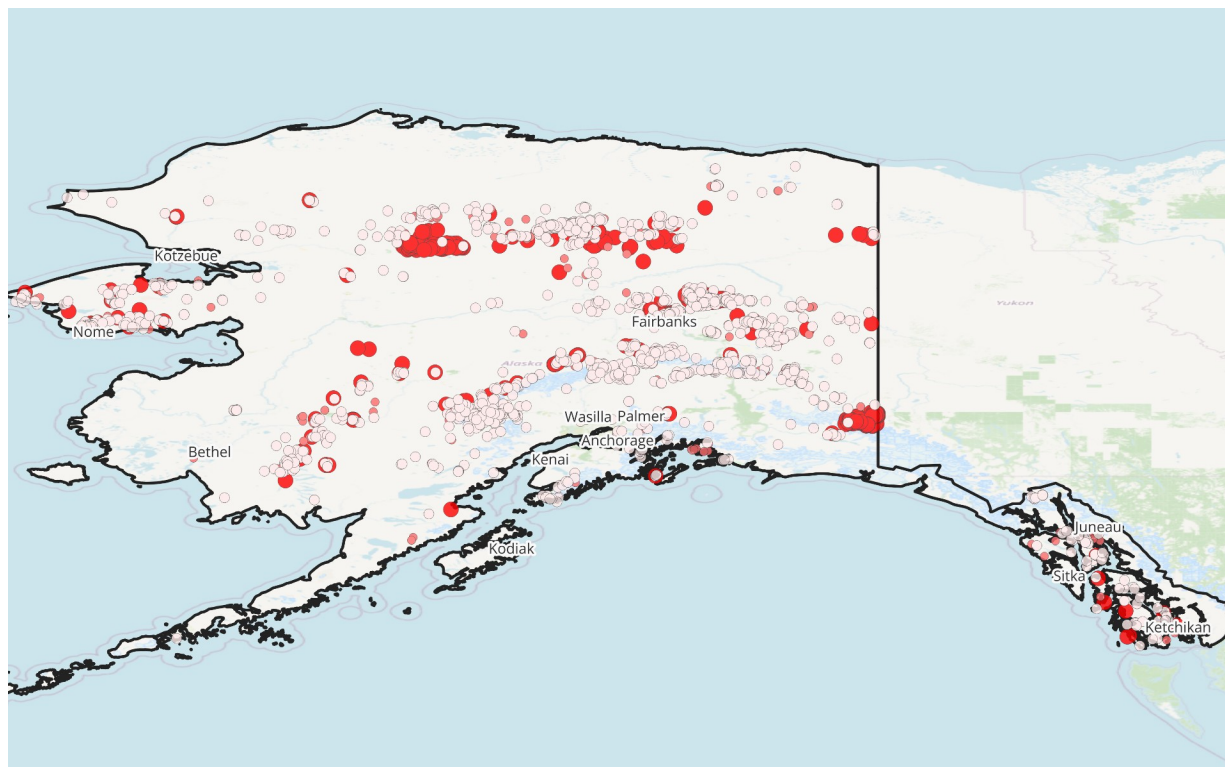
5.3. Geochemical Anomalies

The extraction carried out in the AGDB4 database covered 129,074 stream-sediment samples analyzed for antimony. Of these, 28,968 include a measured value, 91,212 correspond to results below the detection limit, and 8,894 contain no usable value. For measured values, the median is 1.4 ppm, meaning that half of the samples show a concentration below this threshold. The 95th percentile is 14.8 ppm and the 98th percentile is 50 ppm, these two values corresponding respectively to thresholds exceeded by

about 5% and 2% of measured samples. The maximum observed value is approximately 100,000.11 ppm, which illustrates the existence of a few extreme concentrations, far removed from the general distribution of measurements.

From these measurements, 3,110 geochemical anomalies were identified and sorted into three classes: 1,660 moderate anomalies, or about 53.4% of the total, 741 high anomalies, or about 23.8%, and 709 very high anomalies, or about 22.8%. For mapping purposes only, 23 values above 5,000 ppm were visually capped at this threshold, to prevent a very small number of extreme values from making the entire map unreadable; this capping does not modify in any way the values retained in the data.

Map 3 presents the spatial distribution of these 3,110 antimony geochemical anomalies.



Map 3. Antimony Anomalies in Alaska Stream Sediments

Reading note: each point represents a stream-sediment sample showing an antimony concentration considered abnormally high relative to the general distribution of measurements. Three classes are distinguished: moderate anomalies, high anomalies, and very high anomalies, with the most visible symbols corresponding to the strongest anomalies. An abnormally high antimony concentration in a stream sediment may indicate the presence of an enriched source in the rocks or soils of the watershed located upstream of the sampling point. It does not, however, make it possible to precisely locate that source, nor does it alone demonstrate the existence of a deposit. A visual comparison of Maps 1 and 3 suggests a coincidence between certain sectors of high occurrence density and certain sectors of high anomaly density, without any statistical analysis of this association having been carried out as part of this study.

Source: prepared by the author from the USGS Alaska Geochemical Database, version 4. Map background: OpenStreetMap.

5.4. Geochemical Support of Sites

The thirteen documented sites were compared with the presence and intensity of nearby geochemical anomalies, which made it possible to assign them a level of geochemical support among three categories: strong, moderate, or limited. Strong geochemical support is assigned to eight sites: Scrafford (Treasure Creek), Pringle Bench (Jones & Boyle), McCarty (American Eagle), Slate Creek (Taylor mine), Eureka Stibnite (Pick claim group), Hindenburg (Markovich), Stampede, and Upper Ridge. Moderate support is assigned to three sites: Christophosen (Last Chance), Breen West (Bison), and Sliscovich. Limited support is assigned to two sites: Hot Air (Hot Air No. 1) and Quiggley (Gray Eagle). These counts break down across the full set of thirteen documented sites as follows: eight, three, and two.

This level of geochemical support is based on the spatial relationship between each site and the surrounding anomalies; it does not prove that the associated mineralization is continuous, large in volume, or economically workable. One case deserves particular attention: the Upper Ridge site benefits from strong geochemical support, with a geochemical score of 9 based on the available data, but nonetheless obtains a low final priority. Examination of the ranking file indicates that this result is explained by a very low screening score, of only 1 point, associated with a data completeness category indicating that only geological information is available for this site, with no documented grade or tonnage. This case thus illustrates that favorable geochemical support is not enough to guarantee a high final priority when the site-specific information remains too limited.

5.5. Final Ranking

The final ranking combines, for each site, the screening score and the geochemical score, according to the formula presented in Part 4. Seven sites obtain a high final priority: Scrafford (Treasure Creek) and Pringle Bench (Jones & Boyle), both with a score of 25, Slate Creek (Taylor mine) with a score of 24, Eureka Stibnite (Pick claim group) and McCarty (American Eagle), both with a score of 23, then Stampede and Hindenburg (Markovich), both with a score of 21. Five sites obtain a medium final priority: Christophosen (Last Chance) with a score of 18, Breen West (Bison) and Quiggley (Gray Eagle), both with a score of 16, Hot Air (Hot Air No. 1) with a score of 15, and Sliscovich with a score of 14, a value that corresponds to the lower threshold of the medium class. A single site obtains a low final priority: Upper Ridge, with a score of 10.

The Hot Air (Hot Air No. 1) score was recalculated following the unit harmonization presented in Part 4: its documented tonnage, expressed in kilograms, was converted into metric tons, which lowers its tonnage score from 3 to 1 point, its screening score from 12 to 10 points, and its final score from 17 to 15 points. Its screening priority and final priority both remain medium, but its rank drops: it is now ranked after Breen West and Quiggley, tied at 16 points, whereas it previously preceded them.

The ranking includes several score ties, which should be interpreted with caution. Scrafford and Pringle Bench share a score of 25; Eureka Stibnite and McCarty share a score of 23; Stampede and Hindenburg share a score of 21; Breen West and Quiggley share a score of 16. In each of these cases, the presentation order adopted in the table below is a presentation convention and should not be interpreted as demonstrating a difference in potential between the two sites concerned.

Figure 1 presents this ranking as a horizontal bar chart.

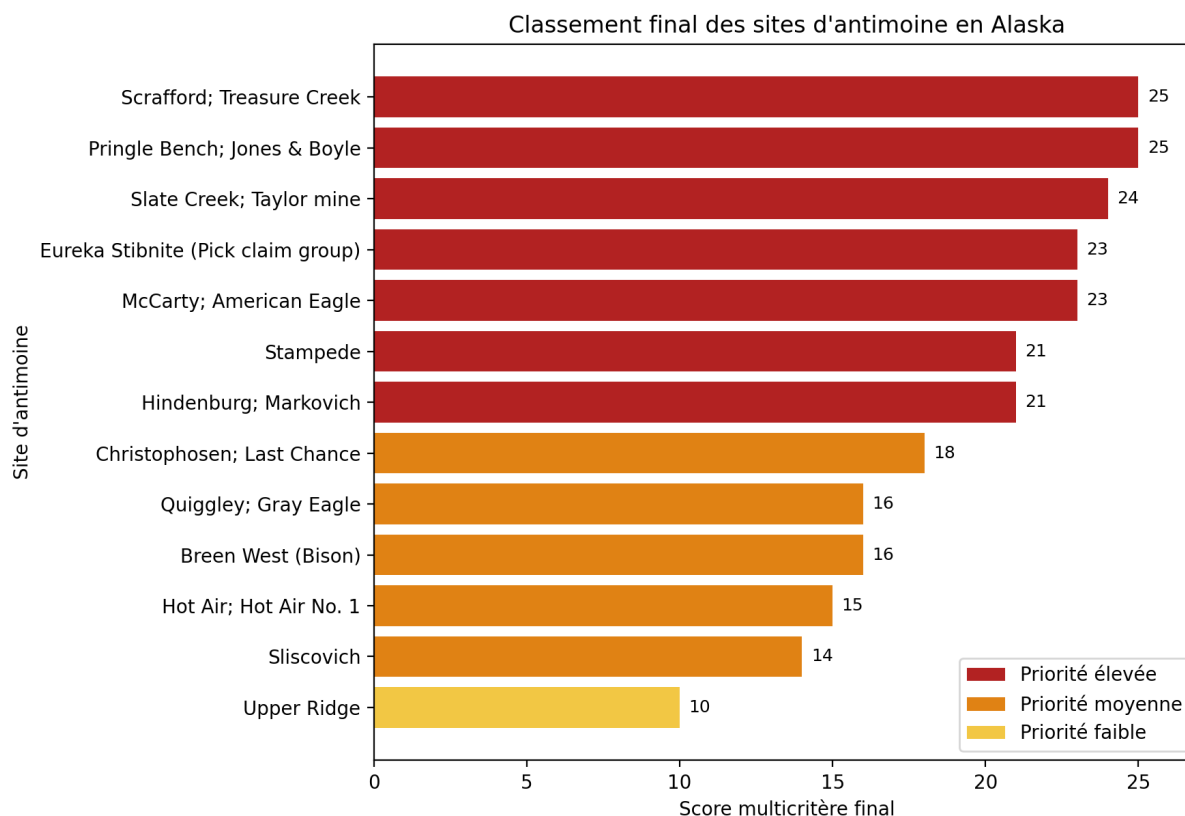
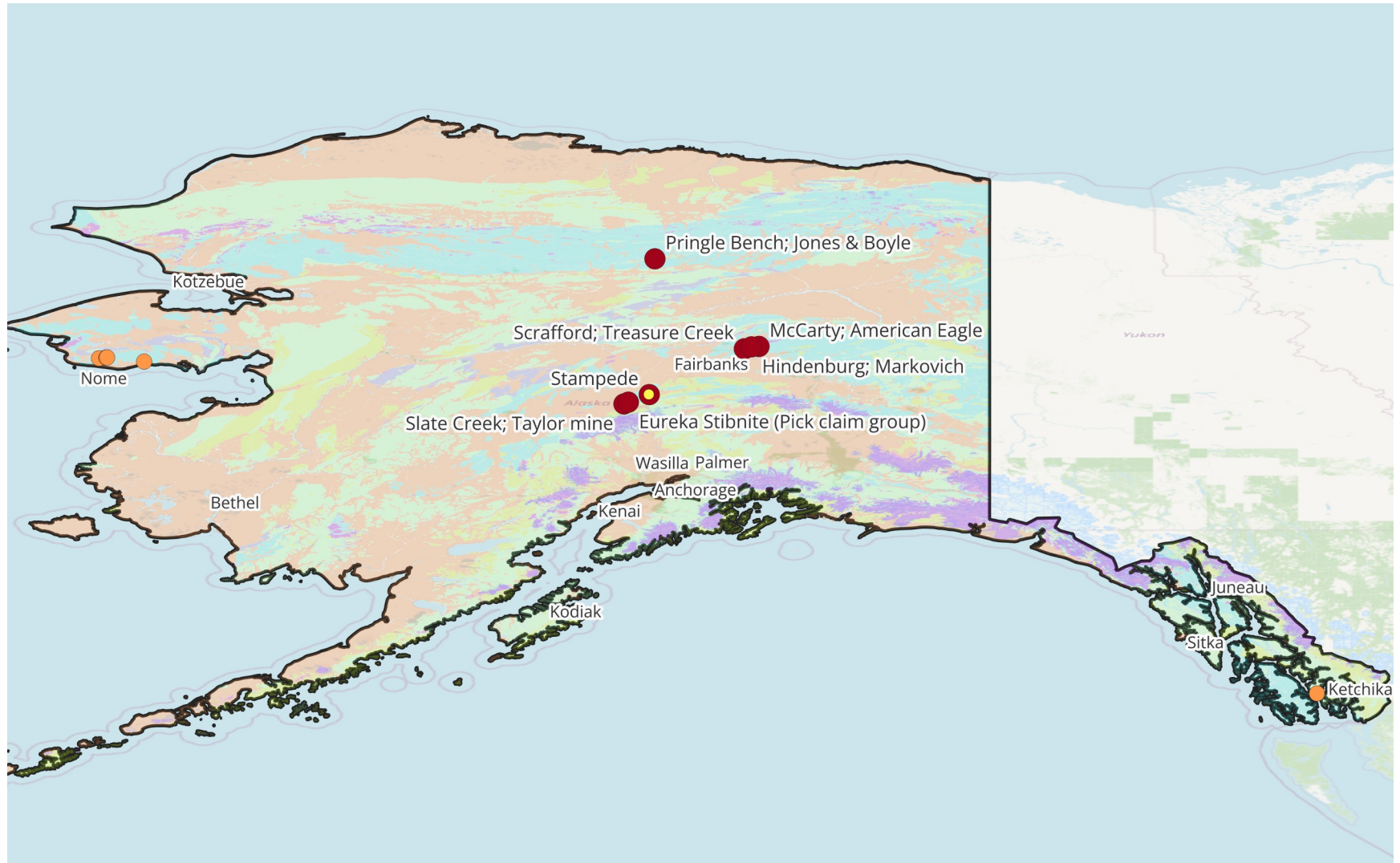


Figure 1. Final Multi-Criteria Ranking of the Antimony Sites Studied in Alaska

Reading note: the length of each bar corresponds to the final score obtained by the site, from a maximum of 25 points to a minimum of 10 points. Sites showing an identical score should be considered tied; their order of appearance in the chart, which for example places Pringle Bench before Scrafford, reflects no demonstrated difference in exploration potential between these two sites.

Source: author's calculations based on the screening and geochemical support scores, antimony_final_ranking.csv file.

Map 4 presents this ranking in its spatial dimension, superimposed on the generalized geological context of Alaska. It is presented in landscape orientation, on a dedicated page, to facilitate reading.



Map 4. Final Antimony Exploration Priorities in Alaska

Reading note: the symbols represent the thirteen sites studied, classified according to their final priority: dark red symbols correspond to high priority, orange symbols to medium priority, and yellow symbols to low priority. The background color represents the major generalized geological

units of Alaska; it provides descriptive regional context and does not directly add points to the final score. The map does not constitute a field geological validation of the ranking presented.

Source: prepared by the author from the USGS Alaska Resource Data File, the USGS Alaska Geochemical Database version 4, and the USGS Scientific Investigations Map 3340. Map background: OpenStreetMap.

Table 4 presents a summary version of the final ranking of the thirteen sites, limited to the essential elements of the score. It is presented in landscape orientation to widen the Site and Grade columns. The qualitative geochemical support level and each site's descriptive geological context appear in Appendix 3, along with all the columns available in the ranking file.

Table 4. Summary of the Final Multi-Criteria Ranking of the Thirteen Documented Sites

Rank	Site	Screening Score	Geochemical Score	Final Score	Final Priority	Grade (% Sb)
1	Scrafford; Treasure Creek	14	11	25	High	55.0
2	Pringle Bench; Jones & Boyle	14	11	25	High	12.78
3	Slate Creek; Taylor mine	14	10	24	High	54.5
4	Eureka Stibnite (Pick claim group)	13	10	23	High	62.0
5	McCarty; American Eagle	12	11	23	High	52.83
6	Stampede	12	9	21	High	12.5
7	Hindenburg; Markovich	11	10	21	High	38.5
8	Christophosen; Last Chance	12	6	18	Medium	58.0
9	Quiggley; Gray Eagle	12	4	16	Medium	63.7
10	Breen West (Bison)	10	6	16	Medium	10.0
11	Hot Air; Hot Air No. 1	10	5	15	Medium	41.03
12	Sliscovich	8	6	14	Medium	35.0
13	Upper Ridge	1	9	10	Low	n.d.

Source: prepared by the author from the `antimony_final_ranking.csv` file, corrected for Hot Air (Hot Air No. 1) after tonnage unit harmonization (see Part 4.2). Note: n.d. means the information is not available in the data provided.

6. Discussion

The approach followed in this report combines two types of information of different natures: on one hand, historical mining archives, which document the presence and sometimes the grade of mineralizations known for a long time, and on the other hand, more systematic regional geochemical data, which make it possible to place these sites within a broader spatial context. This combination provides an element of consistency when the two sources converge, as is the case for most of the high-priority sites, which benefit both from substantial historical documentation and strong geochemical support.

The geographic concentration of occurrences observed on Map 1, notably in north-central Alaska and near Nome, must be interpreted with caution. It may result from regional geology that is indeed more favorable to antimony mineralization, but it may also reflect an uneven history of exploration across sectors of the territory. The most accessible or longest-explored areas, for example near major population centers, may have been more densely documented without this necessarily reflecting geological potential superior to less-documented areas.

The same caution applies to the distinction between well-documented sites, for which a grade or tonnage could be consolidated, and more widely unexplored areas, for which only an occurrence was recorded without associated quantitative information. The absence of numerical data for a sector does not allow one to conclude that potential is absent, but only that sufficient historical documentation is absent.

The value of geochemical anomalies located near sites already known in the ARDF lies in the fact that they provide an independent indicator, obtained through a systematic sampling method rather than through isolated historical documentation. The correspondence observed between several areas of high occurrence density on Map 1 and high anomaly density on Map 3 reinforces the plausibility of a favorable regional context in these sectors, without however constituting direct geological proof.

The limitations relating to grade comparison deserve to be recalled here. The grade values consolidated in this report come from heterogeneous measurement contexts, some corresponding to spot samples and others to historical production estimates. Directly comparing these values as though they had been obtained under a single protocol would lead to a misleading reading of the sites' relative potential. The same applies to tonnages, expressed in different units from one site to another: only those expressed in kilograms or short tons could be converted into metric tons, the others remaining expressed in their original generic unit.

The age of some data is also a factor to take into account. Several descriptions used in the ARDF database date back to work carried out several decades ago, conducted according to methods and documentation standards different from those used today. This age may affect both the precision of the reported grades and the very availability of information for certain sites.

The proposed final ranking also remains sensitive to the scoring rules adopted. The criteria, thresholds, and weightings of this ranking were reconstructed from the Python scripts used in the study and are now presented explicitly in Part 4; the scoring system is therefore reproducible. It nonetheless remains based on threshold and weighting choices specific to this analysis, which have not been subject to any systematic sensitivity analysis: it is not known, at this stage, to what extent small changes to these parameters would

change the ranking obtained. The transparency of the scoring system improves the reproducibility of the ranking, but does not guarantee that the weightings adopted are optimal or universally applicable. This limitation invites the ranking to be considered as an indicative screening tool rather than as a definitive result.

The treatment of tonnage calls for an additional precaution. Values expressed in kilograms or short tons could be converted into metric tons using a precise conversion factor before assigning the score; however, most sites document their tonnage under the sole mention “tons,” without the exact nature of this unit being confirmed in the files provided. These generic tonnages were therefore not converted, for lack of a reliable equivalence, and their tonnage score remains calculated on the raw value. This results in incomplete comparability between sites depending on the unit declared in their original description, a point that cannot be corrected beyond the two cases where the unit was explicit.

The descriptive use of geology in this report also deserves discussion. The geological context associated with each site, whether a metamorphic or metasedimentary context or a context described as other or mixed, comes from a map generalized at the scale of all of Alaska. This regional scale does not make it possible to account for local variations in lithology or structure that may exist within a single site, and should therefore not be interpreted as a precise geological diagnosis of the subsurface.

Finally, it should be recalled that the final score takes into account neither site accessibility, nor existing infrastructure, nor any economic assessment, nor any field validation. The preceding discussion thus invites a clear distinction among several notions: a historically documented site is not necessarily a geochemically supported site; a geochemically supported site is not necessarily a priority site for exploration within the meaning of the proposed ranking; a priority site for exploration is not a deposit in the geological sense of the term; and a deposit is not, in itself, an economically viable project. These distinctions must remain present in the reader's mind throughout the interpretation of the results presented.

7. Recommendations

The recommendations formulated here aim to guide subsequent verification work, in a progressive manner proportionate to the level of uncertainty that characterizes the data used in this report. They do not in any way constitute an invitation to directly undertake exploitation work.

For sites classified as high final priority, namely Scrafford (Treasure Creek), Pringle Bench (Jones & Boyle), Slate Creek (Taylor mine), Eureka Stibnite (Pick claim group), McCarty (American Eagle), Stampede, and Hindenburg (Markovich), the following steps seem reasonable before considering more substantial work:

1. verification of coordinates and site names in the corresponding ARDF records;
2. detailed consultation of the ARDF records and the historical sources cited for each site;
3. verification of the historical sources mentioned in these records;
4. critical reassessment of the reported grades and tonnages, taking into account their original measurement context;
5. local geological mapping, at a finer scale than the generalized map used in this report;
6. field sampling intended to confirm the historical grades;
7. supplementary geochemical campaign, particularly around sites where geochemical support remains limited or moderate;
8. structural study of the local geological context;
9. verification of any former mining works that may have been carried out on these sites;
10. analysis of site accessibility and surrounding infrastructure;
11. review of applicable land-tenure and environmental constraints;
12. supplementary geophysical studies, if they appear justified following the preceding steps;
13. economic study, which should only be undertaken after prior geological validation of the elements above.

For sites classified as medium priority, namely Christophosen (Last Chance), Quiggley (Gray Eagle), Breen West (Bison), Hot Air (Hot Air No. 1), and Sliscovich, an initial documentary verification, supplemented where appropriate by targeted geochemistry in sectors where geochemical support remains limited, appears sufficient at this stage, before considering more costly steps.

The particular case of Upper Ridge, the only site classified as low priority despite strong geochemical support, deserves a specific documentary verification. The absence of documented grade and tonnage for this site, rather than a demonstrated absence of potential, could justify supplementary research in non-digitized archives or in sources external to the ARDF, before any decision regarding the relevance of a field examination.

8. Limitations

This section independently gathers all the limitations that must be kept in mind when interpreting the results presented in this report.

- heterogeneity of the data used, which come from different sources and periods;
- age of several historical sources, established according to documentation standards different from those used today;
- sometimes incomplete descriptions of sites in the ARDF database;
- small number of sites with a consolidated grade, limited to thirteen sites out of the 887 occurrences recorded;
- non-comparable grade contexts, some corresponding to spot samples and others to production estimates;
- limited comparability of tonnages depending on the declared unit: values expressed in kilograms or short tons were converted into metric tons, but most sites mention only a generic “tons” unit, whose exact nature is not confirmed in the files provided and which was therefore not converted;
- missing values for certain sites, notably Upper Ridge, which lacks documented grade and tonnage;
- numerous values below the detection limit in the geochemical data, about 70.7% of the samples analyzed;
- uneven spatial coverage of geochemical sampling across sectors of the territory;
- presence of extreme values in the geochemical data, whose precise origin is not determined in the data provided;
- possible effect of an uneven exploration history on the apparent density of occurrences;
- generalized character of the geological map used, established at a regional scale;
- absence of a subsurface geological model for the full set of sites considered;
- absence of field validation for the results presented in this report;
- absence of any resource or reserve calculation within the meaning of the definitions recalled in Part 1;
- absence of any economic analysis, notably relating to operating costs or site accessibility;
- ranking weightings and thresholds now explicit and reproducible from the scripts, but which remain methodological choices specific to the author, with no systematic sensitivity analysis having been conducted to measure their exact influence on the ranking;
- absence of external validation of the scoring system, which has not been tested against an independent dataset nor submitted to the appraisal of a specialized third party.

Conclusion

This report sought to answer the following question: based on the historical mining occurrences and geochemical data available for Alaska, which sites present the strongest relative interest for further antimony exploration? To answer this, a multi-criteria approach was applied, combining the filtering and harmonization of 887 occurrences from the ARDF database, the consolidation of available grades and tonnages for thirteen documented sites, the extraction and classification of 3,110 geochemical anomalies among 129,074 stream-sediment samples, and the spatial comparison between these anomalies and the recorded sites.

This approach made it possible to establish a final ranking of the thirteen documented sites, sorted into three priority classes. Seven sites obtain a high final priority: Scrafford (Treasure Creek), Pringle Bench (Jones & Boyle), Slate Creek (Taylor mine), Eureka Stibnite (Pick claim group), McCarty (American Eagle), Stampede, and Hindenburg (Markovich). Five sites obtain a medium priority, and a single site, Upper Ridge, obtains a low priority despite strong geochemical support, due to insufficient documentation of its grade and tonnage.

The main interest of this study lies in the cross-referencing of two types of information rarely compared systematically at this scale: historical mining archives, isolated and heterogeneous, and regional geochemical data, more systematic but lacking direct information on the grade or exact nature of any mineralization. The convergence observed, for several sites, between substantial historical documentation and marked geochemical support provides an element of consistency to the analysis.

The ranking methodology is now transparent: the criteria, thresholds, and weightings used to establish the screening score, the geochemical score, and the final score have been made explicit in Part 4 from the study's Python scripts, and the ranking can be reproduced from these rules. This transparency does not, however, change the nature of the report, which remains an exploratory screening analysis: the thresholds and weightings adopted remain methodological choices specific to this study, whose sensitivity has not been systematically tested, and the scoring system has not been subject to any external validation. This report does not in any way constitute a resource or reserve estimate, a feasibility study, an economic assessment, or a field geological validation. Any decision to undertake further work should be preceded by the verification steps presented in Part 7, beginning with detailed documentary consultation and, where appropriate, targeted field sampling at the high-final-priority sites.

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Appendices

Appendix 1. Variables and Fields Used

This appendix presents the full set of variables in the `antimony_final_ranking.csv` file, used to establish the final ranking of the thirteen sites.

Table A1. Variables of the Final Ranking File

Variable	Description
<code>final_rank</code>	Site's rank in the final ranking
<code>site</code>	Site name
<code>final_score</code>	Final score (screening score + geochemical score)
<code>final_priority</code>	Final priority class (high, medium, low)
<code>screening_score</code>	Site-specific screening score
<code>screening_priority</code>	Screening priority class (high, medium, low)
<code>geochemical_score</code>	Score reflecting regional geochemical support
<code>geochemical_support</code>	Qualitative geochemical support level (strong, moderate, limited)
<code>grade_mean_percent</code>	Documented mean antimony grade, as percentage Sb
<code>documented_tonnage</code>	Documented tonnage for the site
<code>tonnage_unit</code>	Unit of the documented tonnage (tons, short tons, kg)
<code>geologic_unit_code</code>	Code of the geological unit associated with the site
<code>geologic_unit_name</code>	Name of the geological unit associated with the site
<code>geologic_age</code>	Geological age of the associated unit
<code>geologic_context</code>	Descriptive geological context
<code>data_completeness</code>	Indicator of data completeness available for the site

Source: prepared by the author from the `antimony_final_ranking.csv` file.

Appendix 2. Ranking System

This appendix recalls the formula and thresholds used to establish the final ranking, the full detail of which appears in Part 4.

- Final score = site's screening score + geochemical score;
- final score of 20 or higher: high final priority; between 14 and 19: medium final priority; below 14: low final priority;
- screening score of 14 or higher: high screening priority; between 10 and 13: medium priority; below 10: low priority;
- geochemical score of 9 or higher: strong geochemical support; between 6 and 8: moderate support; below 6: limited support;
- these three threshold systems are distinct and must not be confused (see Table 3 bis, Part 4.4);

- the screening score adds a grade score (0 to 5 points), a tonnage score (0 to 5 points), a documentary quality score (1 to 5 points), and a quantitative completeness score (0 to 2 points);
- the geochemical score adds a distance-to-nearest-anomaly score (0 to 4 points), a score for the number of very high anomalies within a 25 km radius (0 to 4 points), and a maximum-concentration score within this same radius (0 to 3 points);
- the geological context is included for descriptive purposes and does not directly add points to the final score.

Table A2d below presents, for each of the thirteen sites, the breakdown of the screening score among its four components, reconstructed from the calculation rules of the 05_build_site_screening.py script applied to the grades and tonnages documented in the ranking file. This breakdown is not possible for the geochemical score, whose distance and anomaly-density components do not appear in the final file provided.

Table A2d. Breakdown of the Screening Score by Site

Site	Grade Score	Tonnage Score	Doc. Quality & Completeness	Screening Score
Scrafford; Treasure Creek	5	4	5	14
Pringle Bench; Jones & Boyle	2	5	7	14
Slate Creek; Taylor mine	5	3	6	14
Eureka Stibnite (Pick claim group)	5	2	6	13
McCarty; American Eagle	5	2	5	12
Stampede	2	5	5	12
Hindenburg; Markovich	4	2	5	11
Christophosen; Last Chance	5	1	6	12
Quiggley; Gray Eagle	5	1	6	12
Breen West (Bison)	2	2	6	10
Hot Air; Hot Air No. 1	4	1	5	10
Sliscovich	4	0	4	8
Upper Ridge	0	0	1	1

Source: prepared by the author, breakdown reconstructed from the 05_build_site_screening.py script and from the grade, tonnage, and screening score values in the antimony_final_ranking.csv file. The tonnage score and screening score for Hot Air (Hot Air No. 1) were recalculated after converting its tonnage from 956 kg to 0.956 metric tons (see Part 4.2). The “documentary quality and completeness” column groups the two corresponding components, which cannot be separated from the data provided alone.

Appendix 3. Detailed Ranking of the Thirteen Sites

This appendix reproduces all the columns available in the `antimony_final_ranking.csv` file for each of the thirteen sites. Given the number of variables, the information is split across three complementary tables, presented in landscape orientation to facilitate reading, to be read together by referring to the site name.

Table A2a. Scores and Priorities (Screening, Geochemistry, Final Score)

Rank	Site	Screening Score	Screening Priority	Geochemical Score	Geochemical Support	Final Score	Final Priority
1	Scrafford; Treasure Creek	14	High	11	Strong	25	High
2	Pringle Bench; Jones & Boyle	14	High	11	Strong	25	High
3	Slate Creek; Taylor mine	14	High	10	Strong	24	High
4	Eureka Stibnite (Pick claim group)	13	Medium	10	Strong	23	High
5	McCarty; American Eagle	12	Medium	11	Strong	23	High
6	Stampede	12	Medium	9	Strong	21	High
7	Hindenburg; Markovich	11	Medium	10	Strong	21	High
8	Christophosen; Last Chance	12	Medium	6	Moderate	18	Medium
9	Quiggley; Gray Eagle	12	Medium	4	Limited	16	Medium
10	Breen West (Bison)	10	Medium	6	Moderate	16	Medium
11	Hot Air; Hot Air No. 1	10	Medium	5	Limited	15	Medium
12	Sliscovich	8	Low	6	Moderate	14	Medium
13	Upper Ridge	1	Low	9	Strong	10	Low

Source: prepared by the author from the `antimony_final_ranking.csv` file, corrected for Hot Air (Hot Air No. 1) after tonnage unit harmonization (see Part 4.2).

Table A2b. Grade, Tonnage, and Data Completeness

Site	Grade (% Sb)	Tonnage	Unit	Data Completeness
Scrafford; Treasure Creek	55.0	2,700	tons	Complete screening data
Pringle Bench; Jones & Boyle	12.78	6,308	tons	Complete screening data
Slate Creek; Taylor mine	54.5	354	tons	Complete screening data

Site	Grade (% Sb)	Tonnage	Unit	Data Completeness
Eureka Stibnite (Pick claim group)	62.0	12	tons	Complete screening data
McCarty; American Eagle	52.83	15	tons	Complete screening data
Stampede	12.5	7,000	short tons	Complete screening data
Hindenburg; Markovich	38.5	16.5	tons	Complete screening data
Christophosen; Last Chance	58.0	2.5	tons	Complete screening data
Quiggley; Gray Eagle	63.7	5	tons	Complete screening data
Breen West (Bison)	10.0	14	tons	Complete screening data
Hot Air; Hot Air No. 1	41.03	956	kg	Complete screening data
Sliscovich	35.0	n.d.	n.d.	Grade and geology available
Upper Ridge	n.d.	n.d.	n.d.	Geology only available

Source: prepared by the author from the *antimony_final_ranking.csv* file. Note: n.d. means the information is not available.

Table A2c. Geological Context Associated with Each Site

Site	Geologic Code	Geologic Unit	Geologic Age	Geologic Context
Scrafford; Treasure Creek	DPxsq	Pelitic schist and quartzite (Yukon-Tanana crystalline complex)	Devonian and earlier	Metamorphic or metasedimentary
Pringle Bench; Jones & Boyle	DPxcn	Metasedimentary and metavolcanic rocks of the central belt	Devonian to Proterozoic	Metamorphic or metasedimentary
Slate Creek; Taylor mine	DCsp	Schist and phyllite of the Alaska Range	Devonian or earlier	Metamorphic or metasedimentary
Eureka Stibnite (Pick claim group)	DCsp	Schist and phyllite of the Alaska Range	Devonian or earlier	Metamorphic or metasedimentary
McCarty; American Eagle	DPxsq	Pelitic schist and quartzite (Yukon-Tanana crystalline complex)	Devonian and earlier	Metamorphic or metasedimentary
Stampede	Pzkn	Klondike Schist and Keevy Peak Formation	Permian to early Paleozoic	Metamorphic or metasedimentary
Hindenburg; Markovich	Pze	Eclogite and associated rocks (Yukon-Tanana crystalline complex)	Paleozoic	Other or mixed
Christophosen; Last Chance	PzZncl	Layered sequence (Nome complex)	Devonian to Neoproterozoic	Other or mixed
Quiggley; Gray Eagle	PzZncl	Layered sequence (Nome complex)	Devonian to Neoproterozoic	Other or mixed
Breen West (Bison)	PzZncl	Layered sequence (Nome complex)	Devonian to Neoproterozoic	Other or mixed

Site	Geologic Code	Geologic Unit	Geologic Age	Geologic Context
Hot Air; Hot Air No. 1	CPxwg	Wales Group, undivided	Cambrian to Proterozoic	Other or mixed
Sliscovich	PzZncl	Layered sequence (Nome complex)	Devonian to Neoproterozoic	Other or mixed
Upper Ridge	DCsp	Schist and phyllite of the Alaska Range	Devonian or earlier	Metamorphic or metasedimentary

Source: prepared by the author from the antimony_final_ranking.csv file. Geological ages have been translated into English; geological unit codes and names are reproduced in substance from the original data.

Appendix 4. Geochemical Anomaly Statistics

This appendix gathers the figures relating to stream-sediment samples and antimony geochemical anomalies, presented in the body of the report in Section 5.3. These figures come from the methodological information provided for this study; no detailed tabular file listing each of the 3,110 anomalies individually was provided.

Table A3. Distribution of Stream-Sediment Samples

Category	Count	Share of Total
Measured antimony values	28,968	22.4%
Results below detection limit	91,212	70.7%
Records with no usable value	8,894	6.9%
Total samples analyzed	129,074	100.0%

Table A4. Descriptive Statistics of Measured Values (ppm)

Statistic	Value (ppm)
Median	1.4
95th percentile	14.8
98th percentile	50
Maximum observed value	≈ 100,000.11

Table A5. Classification of Geochemical Anomalies

Anomaly Class	Count	Share of Total
Moderate anomalies	1,660	53.4%
High anomalies	741	23.8%
Very high anomalies	709	22.8%
Total anomalies exported	3,110	100.0%

Source: prepared by the author from the USGS Alaska Geochemical Database, version 4, and the methodological information provided for this study.

Appendix 5. Grade and Tonnage Data

This appendix reproduces, for each of the thirteen documented sites, the mean grade and tonnage as they appear in the ranking file, retained in their original units.

Table A6. Documented Grades and Tonnages by Site

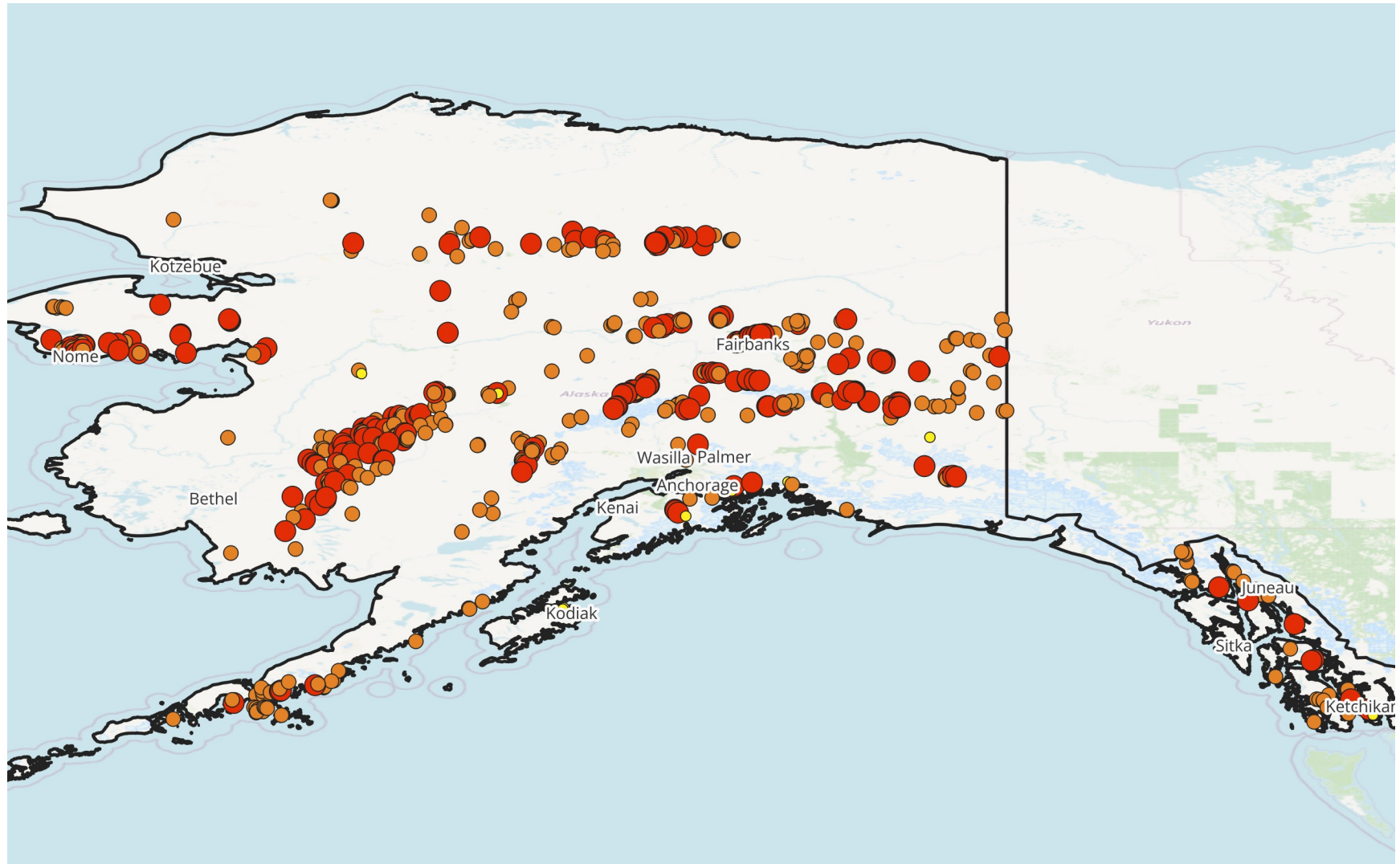
Site	Mean Grade (% Sb)	Documented Tonnage	Unit
Scrafford; Treasure Creek	55.0	2,700	tons
Pringle Bench; Jones & Boyle	12.78	6,308	tons

Site	Mean Grade (% Sb)	Documented Tonnage	Unit
Slate Creek; Taylor mine	54.5	354	tons
Eureka Stibnite (Pick claim group)	62.0	12	tons
McCarty; American Eagle	52.83	15	tons
Stampede	12.5	7,000	short tons
Hindenburg; Markovich	38.5	16.5	tons
Christophosen; Last Chance	58.0	2.5	tons
Quiggley; Gray Eagle	63.7	5	tons
Breen West (Bison)	10.0	14	tons
Hot Air; Hot Air No. 1	41.03	956	kg
Sliscovich	35.0	n.d.	n.d.
Upper Ridge	n.d.	n.d.	n.d.

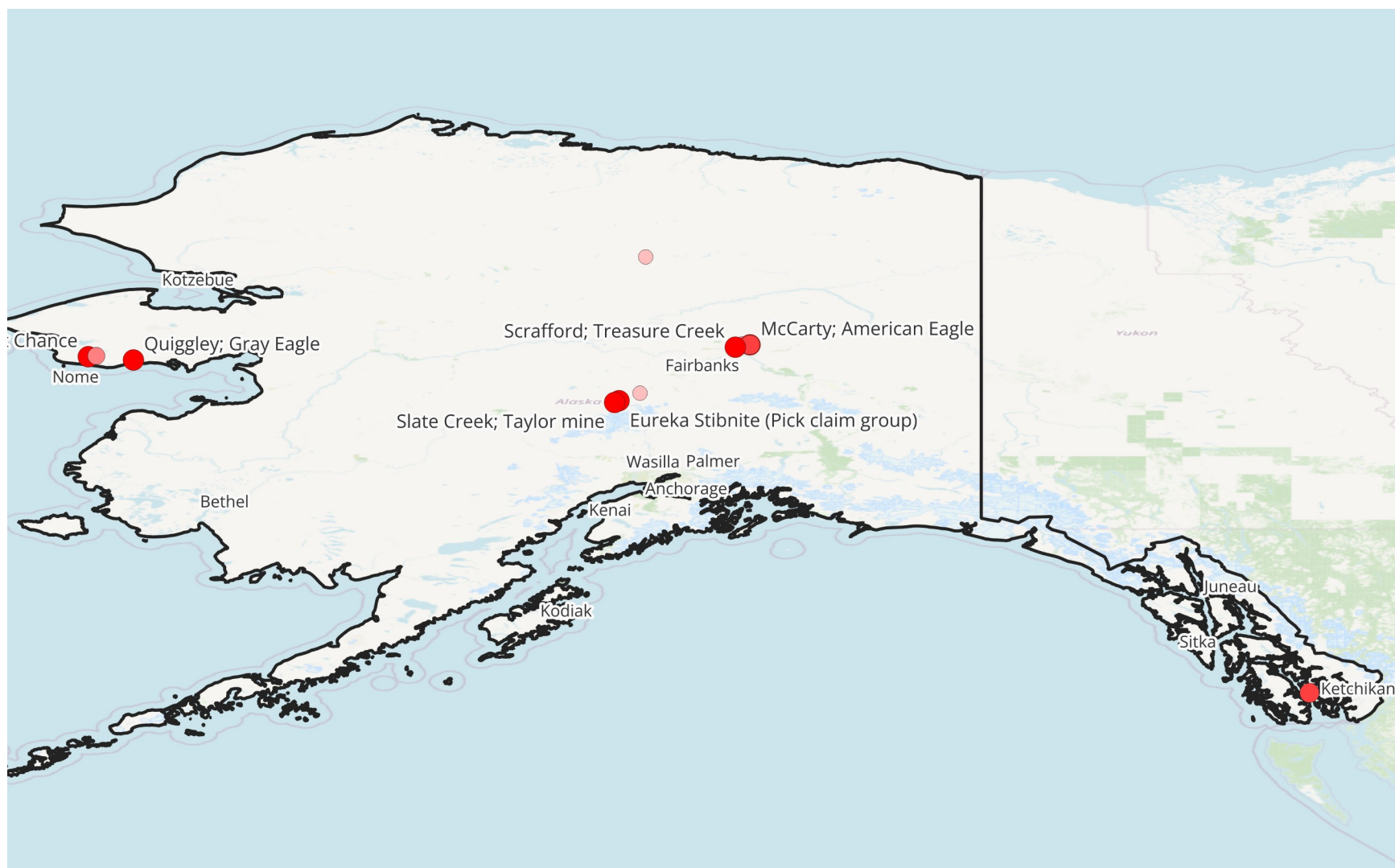
Source: prepared by the author from the *antimony_final_ranking.csv* file. Note: n.d. means the information is not available. Tonnages are presented in their original unit; only those expressed in kilograms or short tons could be harmonized to metric tons for the score calculation (Part 4.2). Tonnages in “tons” remain in an unconfirmed unit and should not be summed across sites.

Appendix 6. Full-Page Maps

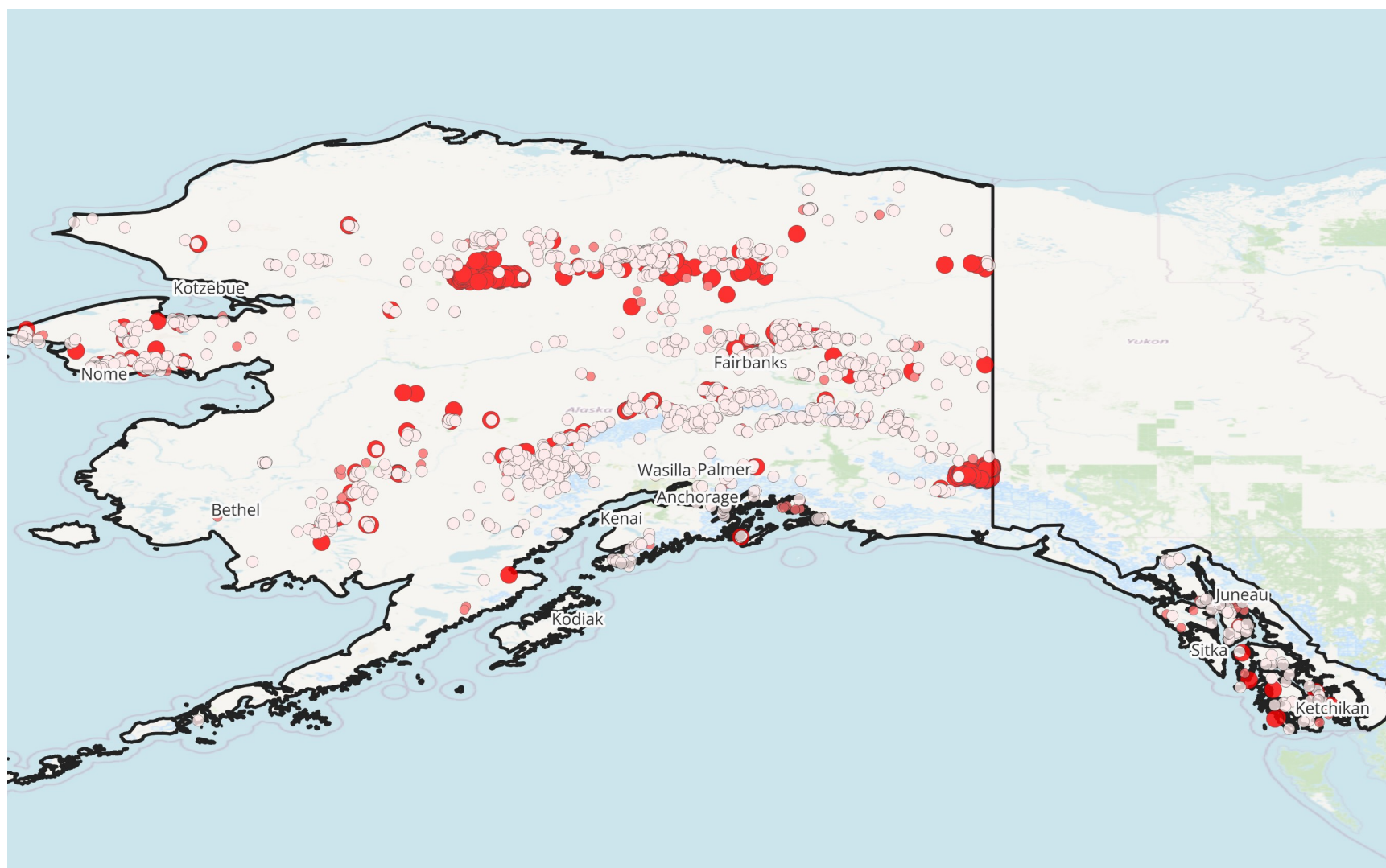
This appendix reproduces, at larger size, in landscape orientation, the four maps presented in the body of the report, to facilitate detailed reading.



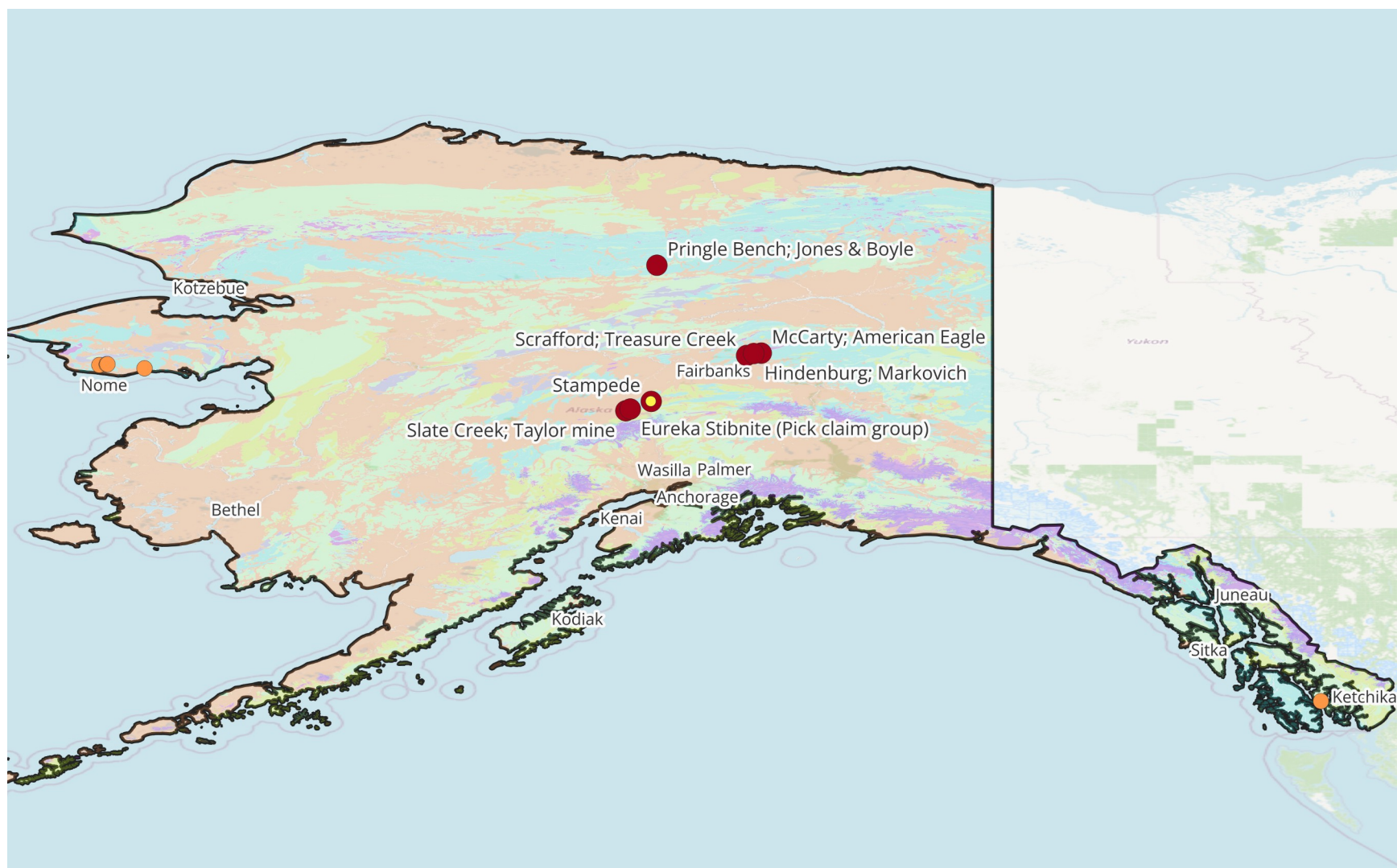
Map 1 (full-page reproduction). Distribution of recorded antimony occurrences in Alaska.



Map 2 (full-page reproduction). Documented antimony grades for the retained sites.



Map 3 (full-page reproduction). Antimony anomalies in Alaska stream sediments.



Map 4 (full-page reproduction). Final antimony exploration priorities in Alaska.